

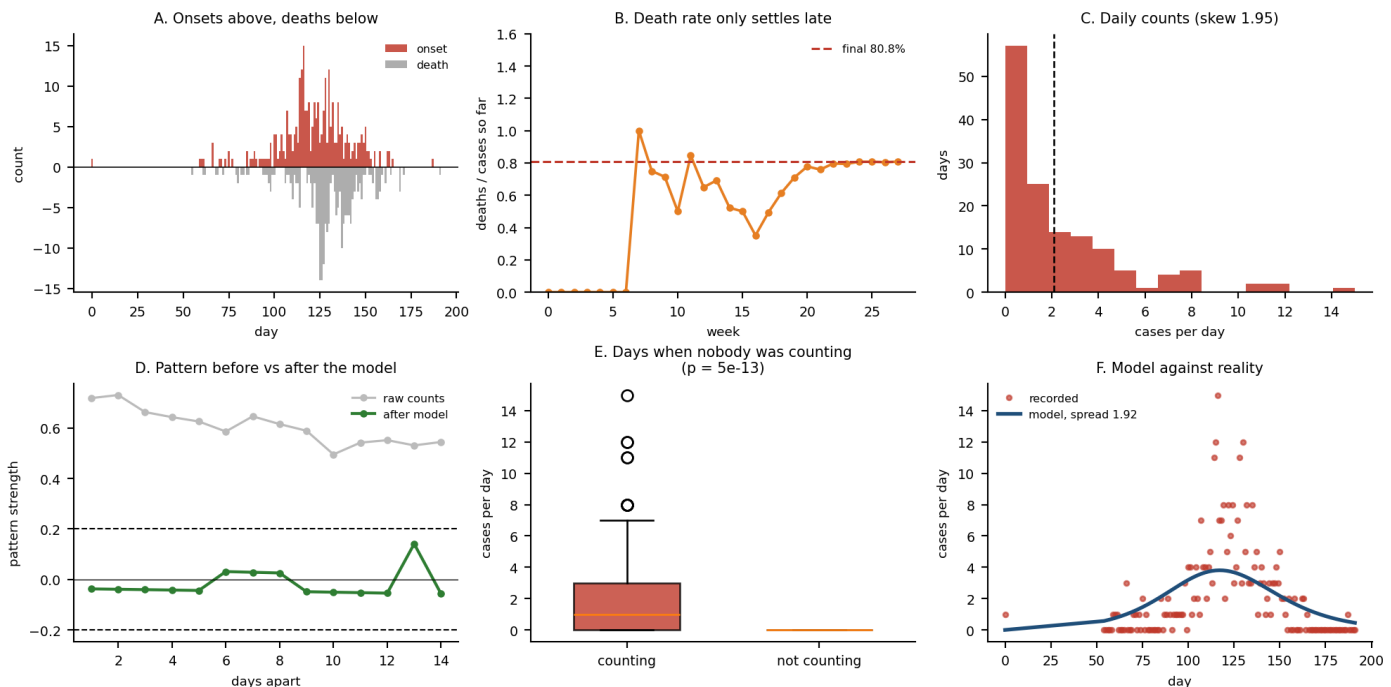
Ebola — Kikwit, DR Congo, 1995

Dossier 1 of 4 · 292 cases over six months · Khan et al. (1999) J Infect Dis 179:S76

Findings, decisions, and where each one stands

Finding	Value in RAW data	Concern	Decision taken	Status in CANONICAL
Many days with zero cases	57.3% of 192 days	could be real zeros or just nobody looking	added analysis_eligible flag; 53 days set aside	RESOLVED — 41.0% of 139 used days
Counts are lopsided	skew 2.41	a plain count model would underestimate the spread	use the forgiving (negative binomial) count	RESOLVED — skew 1.95, model matched to it
Counts jump around	variance/mean 4.63	same concern, measured a second way	same decision	RESOLVED — 4.06
Days follow one another	lag-1 pattern +0.719	suggests something changing over time	v6: compare to NEW CASES per day, not people currently ill	RESOLVED — -0.037 after the model
No-counting days behave differently	chi-square 52.2, $p = 5e-13$	including them would bias everything downward	excluded from the likelihood, kept in the file	RESOLVED — excluded
Deaths lag cases ~14 days	death rate unstable for 10 weeks	day-by-day death rate is meaningless early on	death rate recorded for description only, never modelled	RESOLVED — not a model input

This shows that every concern the raw records raised has a decision attached and evidence that it worked. Nothing here is outstanding.



This shows that panel D is the one to look at: the grey line shows a strong day-to-day pattern in the raw counts, and the green line shows it is gone once the model accounts for the shape of the outbreak. That is what a well-specified model looks like.

What the model will estimate

Quantity	Pre-fit estimate	Published range	Reading
Spread score (R_0)	1.92	1.51 – 2.53 (Althaus 2014)	inside the published range for this same outbreak
Share who died	80.8% (236 of 292)	~81% (Khan 1999)	matches the source publication exactly
Usable observations	139 days	—	second-largest evidence base of the four outbreaks

This shows that the pre-fit estimate already lands where other researchers put it, so the full model has a sensible starting point rather than a surprise to explain.

VERDICT — CLEARED. Six concerns raised by the raw records, six decisions taken, six resolved. No special handling for change-over-time is needed. This outbreak is one of the three carrying the headline comparison.